

Does Sentiment Fusion Help?

A Selection-Bias-Free Re-Evaluation of Multimodal Stock Movement Prediction

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Abstract—Multimodal stock-movement prediction—fusing financial-news and social-media sentiment with technical indicators—is widely reported to deliver large risk-adjusted gains, with recent studies citing average Sharpe ratios above two. We show that such gains are, to a substantial degree, artifacts of optimistic evaluation rather than evidence that sentiment fusion works. We build a selection-bias-free evaluation protocol for multimodal financial prediction—an explicit chronological train/validation/test split with an embargo, model and horizon selection locked to the validation window, and multi-seed reporting with paired significance tests—and apply it to a comprehensive grid of 4,200 task-specific runs (seven model families, eight feature configurations, five U.S. equities, three horizons, five seeds) plus two time-series foundation models (Chronos-2, FinCast). Three findings emerge. *First*, selecting the best model and horizon per stock on the test set inflates the average Sharpe ratio from 1.40 to 2.29—accounting for most of the headline numbers reported in prior work. *Second*, under bias-free evaluation a technical-indicator-only baseline is *not* beaten by any sentiment configuration: fixed 70:30 news–social fusion, equal weighting, news-only, social-only, and a learned reliability-weighted fusion all underperform technical-only, by +0.58 Sharpe on average for fixed 70:30 ($p < 10^{-20}$, paired Wilcoxon). *Third*, time-series foundation models do not beat task-specific models and mostly post negative Sharpe ratios. We conclude that, in this setting, the reported value of multimodal sentiment fusion is largely an evaluation artifact, and we release a reproducible benchmark to support more careful future studies.

Index Terms—stock movement prediction, multimodal fusion, sentiment analysis, evaluation methodology, selection bias, backtest overfitting, financial machine learning, reproducibility

I. INTRODUCTION

Short-term stock-movement prediction is a long-standing application of data mining, and the dominant recent paradigm is *multimodal*: combine quantitative signals (technical indicators derived from price history) with qualitative signals (sentiment extracted from financial news and social media) [1]–[4]. A steady stream of papers reports that incorporating sentiment produces large, economically meaningful gains, often summarized by trading metrics: long/short strategies driven by news sentiment have reported Sharpe ratios above 2.0 net of transaction costs [5].

These reports sit uneasily beside two well-established facts. First, financial return series have a very low signal-to-noise ratio, and the machine-learning finance literature has repeatedly documented how easily backtests *overfit*: with enough models, horizons, and hyperparameter configurations, a strategy that is indistinguishable from noise can be made to look excellent purely by selection [6]–[10]. Second, when general-purpose machine-learning methods are evaluated against simple baselines under disciplined protocols, the reported advantages frequently shrink or disappear [11].

This paper asks a deliberately narrow, falsifiable question: *under a selection-bias-free evaluation protocol, does multimodal sentiment fusion improve short-term stock-movement prediction relative to a technical-indicator baseline?* We answer it with a comprehensive experimental analysis rather than a new model. We re-implement a conventional multimodal-fusion pipeline—FinBERT sentiment, technical indicators, classical and neural classifiers, a long/short backtest—and evaluate it under a protocol designed to control for selection. The apparent advantage of fusion does not survive that protocol.

Concretely, we make three contributions.

- **A selection-bias-free evaluation protocol** for multimodal financial prediction (Section III): an explicit train/validation/test chronological split with an embargo gap; *all* model-family and horizon selection locked to the validation window; multi-seed training; and reporting with paired significance tests alongside backtest-overfitting diagnostics [6], [7].
- **A learned reliability-weighted sentiment fusion** that generalizes the fixed 70:30 news–social heuristic of prior work [12] into a data-driven, prior-regularized weight estimated on validation data (Section III-D). We report it honestly: it does not help.
- **A comprehensive re-evaluation** (Section V) across 4,200 task-specific runs and two time-series foundation models, yielding three insights: (i) test-set selection inflates the average Sharpe ratio from 1.40 to 2.29; (ii) no sentiment configuration beats a technical-only baseline under bias-free evaluation; (iii) foundation models do not beat task-specific models. We release code and the

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processed benchmark at GitHub¹.

Our intent is not to claim that sentiment is uninformative in all settings, but to show that a frequently reported positive result does not survive careful evaluation—a comprehensive experimental analysis that yields new insight into the strengths and weaknesses of an existing and widely used method. Fig. 1 summarizes the study: a conventional multimodal pipeline, executed inside a selection-bias-free evaluation protocol.

II. RELATED WORK

Multimodal stock-movement prediction. Combining textual sentiment with market data is now standard. StockNet [1] pairs tweets with prices; MAN-SF [2] adds attentive cross-modal and inter-stock modeling; more recent work fuses news, social media, and engineered indicators [3], [4], [13]. Domain-adapted language models such as FinBERT [14], [15] are the usual sentiment extractors. A recurring design choice is how to weight news against social media: Nyakurukwa and Seetharam [12] report that news sentiment correlates more strongly with returns, motivating a fixed 70:30 news-to-social weighting that several pipelines adopt. Most of these studies report sizeable gains for fusion over price- or indicator-only baselines.

Time-series foundation models. General-purpose forecasters pre-trained on large, heterogeneous corpora—e.g., Chronos-2 [16] and the finance-specific FinCast [17]—offer zero-shot and fine-tuned prediction and can, in principle, ingest covariates. Whether their broad pre-training transfers to the noisy, non-stationary regime of equity returns is not well characterized.

Evaluation pitfalls in financial ML. The financial-machine-learning literature has long warned that the binding constraint is evaluation, not modeling. López de Prado [8] catalogues leakage and overfitting failure modes; Bailey and López de Prado [6] introduce the Deflated Sharpe Ratio to correct an observed Sharpe ratio for the number of trials conducted; the Probability of Backtest Overfitting [7] quantifies how often an in-sample-best configuration underperforms out of sample; Harvey et al. [10] and Arnott et al. [9] document multiple-testing inflation across the field. Empirical-asset-pricing studies that adopt disciplined protocols [18] typically report modest, carefully bounded effects. Our contribution is to bring this discipline to bear specifically on the *multimodal-fusion* question, and to quantify how much of the reported advantage it removes.

III. METHODOLOGY AND EVALUATION PROTOCOL

We first state the prediction task, then describe the four elements of the protocol that distinguish bias-free evaluation from the optimistic evaluation common in prior work.

A. Problem Formulation

For a stock s with closing-price series $\{p_t\}$, the binary movement label at horizon h is

$$y_{t+h} = \mathbb{I}\left[\frac{p_{t+h} - p_t}{p_t} > \theta\right], \quad (1)$$

where $\mathbb{I}[\cdot]$ is the indicator function and θ is a per-(stock, horizon) threshold chosen *on the training window* to balance the two classes. A model receives a feature vector $\mathbf{z}_t$ combining technical indicators $\mathbf{X}_t$ and an aggregated daily sentiment score S_t (Section IV), and outputs $\hat{p}(y_{t+h} = 1)$. Predictions are evaluated both as a classification problem (ROC-AUC) and through a long/short trading simulation with non-overlapping h -day positions and 10 bps round-trip transaction costs.

B. Chronological Split with Embargo

All data span January 2020 to December 2024. We use a strict calendar split: **training** 2020–2022, **validation** 2023, **test** 2024. The last h rows of the validation window are dropped so that validation labels, which look h days ahead, cannot reach into the test year, and a further five-trading-day *embargo* is removed from the start of the test window [8]. This replaces the train/test-only split of prior pipelines, which leaves no clean data on which to make modeling choices.

C. Validation-Locked Model and Horizon Selection

This is the single most consequential element. Prior multimodal pipelines typically report, for each stock, the *best* model family and prediction horizon—but the “best” is identified on the test year. With seven model families and nine candidate horizons, that is up to 63 trials per stock; the maximum over 63 noisy trials is a badly upward-biased estimate of true performance [6], [10].

We instead compute, for every (stock, ablation, model, horizon) cell, the *validation* ROC-AUC and validation Sharpe ratio, and write the selection—which model and which horizon—to an immutable configuration file keyed only by validation scores. Every number reported on the 2024 test year is then read back through that file. No test-set quantity is ever used to make a modeling choice. Section V-A quantifies how much this single change removes.

D. Learned Reliability-Weighted Fusion

Prior work aggregates daily news sentiment S_t^{news} and social sentiment S_t^{soc} with fixed weights [12]:

$$S_t = 0.7 S_t^{\text{news}} + 0.3 S_t^{\text{soc}}, \quad (2)$$

We generalize Equation 2 by introducing a learnable weight:

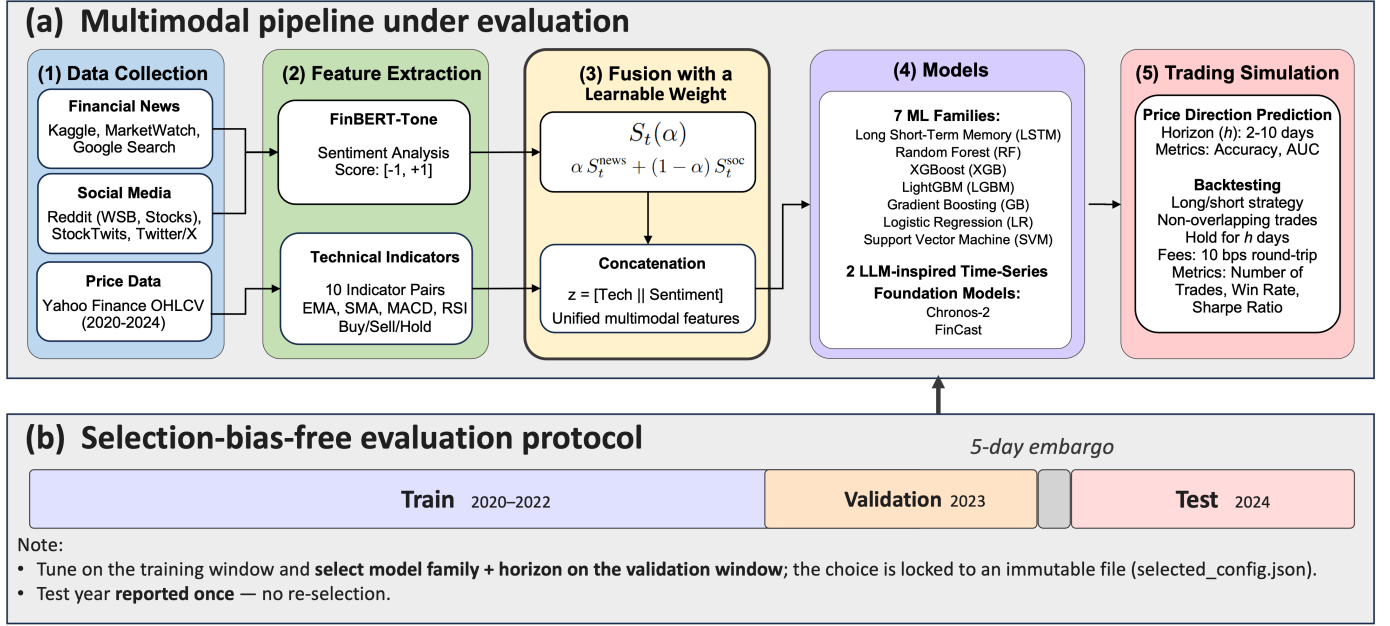
$$S_t(\alpha) = \alpha S_t^{\text{news}} + (1 - \alpha) S_t^{\text{soc}} \quad (3)$$

where $\alpha \in [0, 1]$ is a learnable weight estimated by maximizing the validation ROC-AUC with a quadratic penalty pulling α toward the literature value of 0.7:

$$\alpha^* = \arg \max_{\alpha} \text{AUC}_{\text{val}}(\alpha) - \lambda (\alpha - 0.7)^2. \quad (4)$$

The fixed 70:30 rule is recovered as $\lambda \rightarrow \infty$. We fit α both per stock and globally (one shared α). This makes the weighting a data-driven choice that the evaluation can test rather than assume.

¹<https://github.com/inflaton/Multimodal-Stock-Movement-Prediction>



Grid: 8 feature configurations $\times$ 7 model families $\times$ 3 horizons $\times$ 5 seeds = **4,200 task-specific runs** (plus 2 time-series foundation models). Reported as multi-seed mean $\pm$ std with paired Wilcoxon signed-rank tests.

Fig. 1. Overview of the study. (a) A conventional multimodal stock-movement pipeline: news, social-media, and price data are turned into FinBERT-Tone sentiment and technical-indicator features, fused, fed to seven model families, and evaluated by a long/short backtest. (b) The selection-bias-free evaluation protocol that is the methodological contribution: a chronological train/validation/test split with a five-day embargo; *all* model-family and horizon selection locked to the 2023 validation window; and a 4,200-run multi-seed grid reported with paired significance tests. Every grid cell is one execution of pipeline (a).

E. Multi-Seed Reporting and Statistics

Every stochastic configuration is trained with five random seeds; we report the mean and standard deviation across seeds. Because the central question is a *paired* comparison—does configuration A beat configuration B on the same stock, model, horizon, and seed?—we use the Wilcoxon signed-rank test over all matched cells, which makes no normality assumption. We additionally adopt the Deflated Sharpe Ratio [6] and the Probability of Backtest Overfitting [7] as backtest-overfitting diagnostics; their implementations are included in the released code. Trading metrics reported are the Sharpe ratio [19], win rate, cumulative return over the test year, and maximum drawdown (MDD).

IV. EXPERIMENTAL SETUP

A. Data

We study five large, liquid U.S. equities spanning distinct sectors and volatility regimes: AAPL, META, NVDA, SPY, and TSLA. Daily OHLCV prices are from Yahoo! Finance [20]. Financial-news and social-media text (2020–2024) is scored with FinBERT-Tone [15] to produce a per-document sentiment in $[-1, 1]$, then averaged per trading day into S_t^{news} and S_t^{soc} . Table I summarizes coverage. Days lacking text of a given type are imputed as neutral (0); such days are rare except for SPY news, which we revisit in Section VII. A first warning sign is visible already here: the correlation between

TABLE I
DATASET COVERAGE, 2020–2024 (1,257 TRADING DAYS PER STOCK).
“COV.” IS THE FRACTION OF TRADING DAYS WITH AT LEAST ONE
DOCUMENT OF THAT TYPE.

Stock	News cov.	Social cov.	Mean S_t^{news}	Mean S_t^{soc}
AAPL	0.993	0.996	0.456	0.690
META	0.936	0.887	0.522	0.803
NVDA	0.937	0.994	0.455	0.665
SPY	0.835	0.947	0.371	0.767
TSLA	1.000	0.991	0.473	0.752

daily news sentiment and next-day return is tiny and sign-inconsistent across stocks (between -0.05 and $+0.06$).

B. Feature Configurations (Ablations)

The technical branch follows prior work [4]: ten indicator-pair signals per stock (moving-average, MACD, RSI, and Bollinger-band crossovers), each encoded as Buy/Sell/Hold $\in \{+1, -1, 0\}$ and selected on the training window only. We evaluate eight feature configurations: **technical_only** (indicators, no sentiment); **full_70_30** (indicators + fixed 70:30 fusion); **equal_50_50** (indicators + 50:50 fusion); **news_only** and **social_only** (indicators + single-source sentiment); **sentiment_only** (fused sentiment, *no* indicators); **learned_alpha_per_stock** and **learned_alpha_global** (indicators + the learned fusion of Section III-D).

TABLE II
SELECTION BIAS: BEST (MODEL, HORIZON) PER STOCK FOR THE
FIXED-70:30 PIPELINE, SELECTED ON THE TEST SET VS. THE VALIDATION
SET. TEST-YEAR SHARPE RATIO AND ROC-AUC.

Stock	Test-peek selection		Validation-locked	
	Sharpe	AUC	Sharpe	AUC
AAPL	0.55	0.529	-0.45	0.533
META	1.89	0.586	1.89	0.586
NVDA	4.14	0.599	4.14	0.599
SPY	2.33	0.571	1.11	0.607
TSLA	2.57	0.514	0.30	0.502
Average	2.29	0.560	1.40	0.565

C. Models

Task-specific. Seven families: Logistic Regression (LR), SVM, Random Forest [21], Gradient Boosting, XGBoost [22], LightGBM [23], and a two-layer LSTM [24]. Each is tuned with 30 iterations of Bayesian optimization [25] on the validation window. **Foundation.** Chronos-2 [16] in four configurations (zero-shot and fine-tuned, each with and without sentiment/indicator covariates) and FinCast [17], whose released artifact supports only price-only inputs.

D. Grid and Implementation

The task-specific grid is 7 models $\times$ 8 ablations $\times$ 3 horizons $\{2, 5, 10\} \times 5$ seeds $\times 5$ stocks = 4,200 runs; foundation models add 75 runs. Backtesting uses a long/short strategy, non-overlapping h -day positions, and 10 bps round-trip costs.

V. RESULTS AND ANALYSIS

A. Selection Bias Inflates the Headline

We first isolate the effect of test-set selection. For the conventional fixed-70:30 pipeline, we select the best (model, horizon) per stock two ways: *test-peek*, maximizing the 2024 test Sharpe ratio (the implicit procedure behind many reported numbers), and *validation-locked*, maximizing the 2023 validation Sharpe ratio and then reading the test number back. Table II shows the result. Test-peek selection yields a 2.29 average Sharpe ratio—squarely in the range prior work reports—while the identical pipeline, selected honestly, yields 1.40. Roughly 0.9 units of Sharpe ratio, about 40% of the headline, is pure selection bias. Discrimination is barely affected (AUC 0.560 vs. 0.565): the bias lives almost entirely in the trading metric, exactly where overfitting concentrates [6].

B. Does Sentiment Fusion Help? No.

The central experiment compares the eight feature configurations under bias-free evaluation. Table III reports the mean over the entire grid—all seven model families, three horizons, five stocks, five seeds (525 runs per configuration)—which removes any dependence on a hand-picked model or horizon.

A note on aggregation, since two views of the data appear in this paper. Table II reports the *best configuration per stock*—one model and one horizon chosen for each stock—which

is why its honest, validation-locked figure for fixed 70:30 is 1.40. Table III instead reports the *grid mean*: the average over every model and horizon with no selection at all, 0.23 for the same configuration. These are not in conflict—one is a favourable per-stock pick, the other an unselected average—and the grid mean is the unbiased estimate of a configuration’s worth. Table IV reports both views for all eight configurations: **technical_only** ranks first under *both* the grid mean and the per-stock best-configuration view, while the ordering of the sentiment configurations is itself unstable between the two—per-stock selection inflates every configuration and reshuffles their ranks, whereas the grid mean exposes the decisive gaps. Sentiment loses regardless of how performance is aggregated.

The result is unambiguous. **technical_only** attains the highest Sharpe ratio (0.81), the highest AUC (0.542), the highest win rate, and the shallowest drawdown. *Every* configuration that adds sentiment does worse, and the gap is statistically decisive: the paired Wilcoxon test rejects equality at $p < 10^{-7}$ for all seven comparisons, with technical-only winning 57–70% of the 525 matched cells. Adding fixed 70:30 fusion costs 0.58 Sharpe units on average; replacing indicators entirely with sentiment (**sentiment_only**) costs 1.11 and drives both AUC and win rate below chance. The controlled single-model view (LightGBM at $h = 5$) tells the same story—technical-only 0.95 Sharpe vs. 0.23 for full_70_30—so the finding is not an artifact of pooling.

This directly contradicts two claims common in the multimodal-fusion literature: that fusion is “essential” and that sentiment “drives profitability.” Under bias-free evaluation, sentiment is, at best, an inert feature and, at worst, a source of noise that the downstream classifier cannot discount.

C. The Learned Fusion Does Not Rescue It

The learned reliability weight (Section III-D) was our intended methodological improvement. It behaves sensibly—Table V shows α^* landing near or above the literature value of 0.7 for four of five stocks—but it does not change the conclusion. On validation AUC it beats the fixed 70:30 rule by a negligible margin for four stocks ($\Delta \leq 0.005$); only META shows a real gain (+0.047). On the test year, **learned_alpha_per_stock** (0.39 Sharpe) and **learned_alpha_global** (0.21) still lose to technical-only by 0.42 and 0.61 Sharpe units respectively (Table III, $p < 10^{-9}$). A better-fit weight on a feature that carries no signal yields a better-fit version of nothing.

D. Model Families: Simpler Is Stronger

Table VI breaks the task-specific grid down by model family (each row averaged over all eight feature configurations, three horizons, five stocks, and five seeds). The ordering is itself informative: Logistic Regression—the simplest, most heavily regularized family—is the strongest on every metric (0.547 AUC, 0.90 Sharpe), while the highest-capacity families, the LSTM and the SVM, are the weakest and post negative Sharpe ratios. In a regime where the dominant feature is a sparse set of Buy/Sell/Hold indicator signals and the textual features carry

TABLE III

FEATURE-CONFIGURATION COMPARISON UNDER BIAS-FREE EVALUATION. EACH ROW IS THE MEAN OVER 525 RUNS (7 MODELS $\times$ 3 HORIZONS $\times$ 5 STOCKS $\times$ 5 SEEDS); SHARPE SHOWN AS MEAN $\pm$ STD. Δ SHARPE AND p ARE THE PAIRED WILCOXON COMPARISON OF **TECHNICAL_ONLY** MINUS THAT ROW, OVER ALL 525 MATCHED CELLS. POSITIVE Δ MEANS TECHNICAL-ONLY WINS.

Configuration	Test AUC	Test Sharpe	Win rate	Cum. return	MDD	Δ Sharpe vs. tech.	p (Wilcoxon)
technical_only	0.542	0.81 $\pm$ 1.61	0.542	0.73	−0.265	—	—
equal_50_50	0.536	0.46 $\pm$ 1.45	0.506	0.45	−0.275	+0.35	1.1×10^{-7}
learned_alpha_per_stock	0.532	0.39 $\pm$ 1.40	0.508	0.38	−0.272	+0.42	3.6×10^{-10}
news_only	0.541	0.24 $\pm$ 1.39	0.504	0.33	−0.283	+0.58	6.4×10^{-20}
full_70_30	0.532	0.23 $\pm$ 1.46	0.496	0.37	−0.295	+0.58	2.4×10^{-21}
learned_alpha_global	0.530	0.21 $\pm$ 1.51	0.495	0.37	−0.295	+0.61	1.9×10^{-15}
social_only	0.515	−0.05 $\pm$ 1.33	0.482	0.09	−0.322	+0.86	4.5×10^{-30}
sentiment_only	0.499	−0.30 $\pm$ 1.18	0.469	−0.06	−0.346	+1.11	4.0×10^{-31}

TABLE IV

THE HEADLINE RANKING IS INVARIANT TO AGGREGATION. *Grid-mean Sharpe*: MEAN TEST SHARPE OVER ALL 525 RUNS OF A CONFIGURATION (THE TEST SHARPE COLUMN OF TABLE III). *Best-per-stock Sharpe*: FOR EACH STOCK THE (MODEL, HORIZON) WITH THE HIGHEST *validation* SHARPE IS SELECTED AND ITS TEST SHARPE RECORDED; THE FIVE ARE AVERAGED. **TECHNICAL_ONLY** RANKS FIRST UNDER BOTH; PER-STOCK SELECTION INFLATES EVERY CONFIGURATION AND RESHUFFLES THE REST.

Configuration	Grid-mean Sharpe	Best-per-stock Sharpe
technical_only	0.81	2.26
equal_50_50	0.46	1.13
learned_alpha_per_stock	0.39	2.06
news_only	0.24	1.20
full_70_30	0.23	1.40
learned_alpha_global	0.21	2.11
social_only	−0.05	1.20
sentiment_only	−0.30	0.16

TABLE V

LEARNED NEWS WEIGHT α^* VS. THE FIXED 0.7 HEURISTIC, AND THE VALIDATION-AUC GAIN. ESTIMATED ON 2023 VALIDATION DATA WITH PRIOR STRENGTH $\lambda = 0.1$.

Stock	α^*	Val AUC (learned)	Δ vs. $\alpha=0.7$
AAPL	0.72	0.563	+0.001
META	0.84	0.618	+0.047
NVDA	0.82	0.656	+0.005
SPY	0.80	0.577	+0.004
TSLA	0.54	0.585	+0.004
Global	0.80	0.598	+0.011

almost no signal, additional model capacity mostly fits noise. This is the same lesson as the feature ablation, seen from the model side: when the data are this noisy, restraint—in features and in capacity—outperforms expressiveness.

E. Foundation Models Do Not Beat Task-Specific Models

Table VII compares the five foundation-model configurations against the strongest task-specific family. Every foundation configuration posts a near-chance AUC (0.47–0.50) and, in aggregate, a negative Sharpe ratio. Fine-tuning helps

TABLE VI

TASK-SPECIFIC MODEL FAMILIES, EACH AVERAGED OVER THE FULL FEATURE $\times$ HORIZON $\times$ STOCK $\times$ SEED GRID (600 RUNS PER FAMILY).

Model family	Test AUC	Test Sharpe	Win rate
Logistic Regression	0.547	0.90	0.542
Random Forest	0.539	0.26	0.494
XGBoost	0.533	0.36	0.503
SVM	0.531	−0.06	0.478
Gradient Boosting	0.528	0.11	0.487
LightGBM	0.519	0.24	0.501
LSTM	0.499	−0.04	0.497

TABLE VII

FOUNDATION MODELS VS. THE STRONGEST TASK-SPECIFIC FAMILY. FOUNDATION ROWS ARE MEANS OVER 15 RUNS (5 STOCKS $\times$ 3 HORIZONS, SINGLE SEED); THE TASK-SPECIFIC ROW IS THE MEAN OVER ITS FULL GRID.

Method	Test AUC	Test Sharpe	Win rate
Chronos-2 zero-shot	0.472	−0.38	0.454
Chronos-2 zero-shot + cov.	0.482	−0.61	0.436
Chronos-2 fine-tuned	0.497	−0.54	0.450
Chronos-2 fine-tuned + cov.	0.484	−0.02	0.470
FinCast (price-only) [†]	0.498	−0.70	0.425
Task-specific (LR)	0.547	0.90	0.542

[†]The released FinCast artifact does not accept covariates; this is an artifact limitation, not a property of the model class.

Chronos-2, and adding covariates helps further—the fine-tuned-with-covariates variant is the only one to approach break-even (−0.02 Sharpe)—which is consistent with the view that the gap to task-specific models narrows under matched information and tuning. But none beats the task-specific models: the best task-specific family (Logistic Regression) reaches 0.90 average Sharpe and 0.547 AUC. The often-quoted claim that “lightweight task-specific models beat foundation models” thus holds in our data—but it is a weak consolation, because the task-specific “win” is itself driven by technical indicators, not by the multimodal fusion the foundation models were meant to be contrasted against.

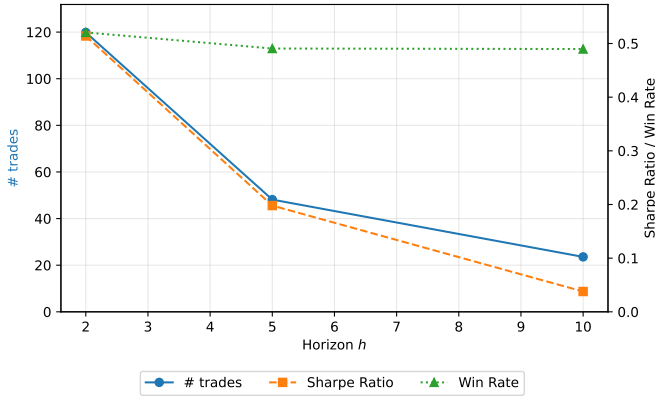


Fig. 2. Mean number of trades, Sharpe ratio, and win rate by horizon over the full grid. Predictability concentrates at the shortest horizon and decays.

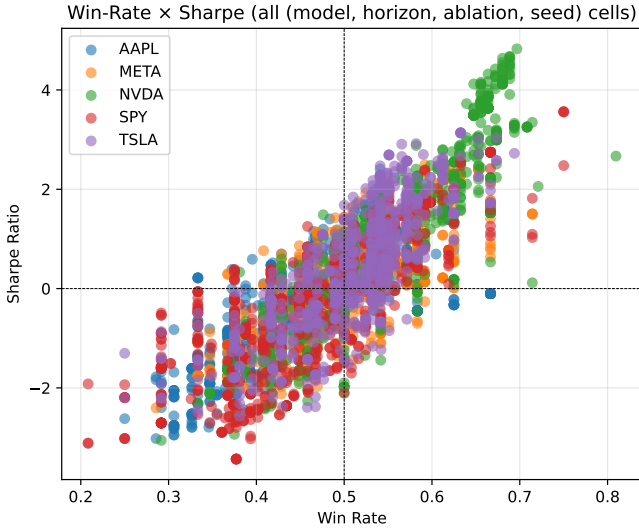


Fig. 3. Win rate vs. Sharpe ratio across all task-specific runs. The two are only weakly related: a win rate above 0.5 does not guarantee a positive Sharpe ratio once transaction costs and return asymmetry are accounted for.

F. Where Any Signal Lives: Horizon and Discrimination

Two secondary patterns help interpret the main result. First, what little predictability exists is concentrated at the shortest horizon: averaged over the grid, $h = 2$ gives 0.51 Sharpe and 0.527 AUC, decaying to 0.04 Sharpe at $h = 10$ (Fig. 2). Second, win rate and Sharpe ratio are only loosely coupled across configurations (Fig. 3): many configurations clear a 50% win rate yet still lose money after costs, because a few large adverse moves dominate. Both patterns are consistent with a very low underlying signal-to-noise ratio in which evaluation choices, not features, determine the headline.

VI. DISCUSSION

Why does fusion fail here? The proximate cause is signal: daily FinBERT-Tone sentiment correlates with next-day returns at $|\rho| \leq 0.06$, with inconsistent sign across stocks.

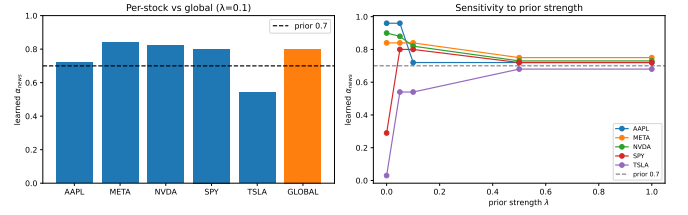


Fig. 4. Learned news weight α^* per stock (and global), and its sensitivity to the prior strength λ . The weights are stable and near the 0.7 heuristic, but the configurations built on them still lose to technical-only (Table III).

A near-zero-signal, noisy regressor concatenated onto an informative technical-indicator vector cannot improve a well-regularized classifier and can degrade it—which is exactly the ordering in Table III. Aggregating thousands of documents into one daily scalar discards most of the textual content; the scalar that remains is dominated by a persistently positive tone (Table I) that varies little with the next move.

Why is this not visible in prior work? Two reasons. First, test-set selection (Table II) manufactures roughly 0.9 Sharpe units regardless of features; a pipeline that quotes the per-stock maximum will look strong whether or not sentiment helps. Second, isolated-modality ablations are typically read on the trading metric alone, which is dominated by variance at small trade counts; a careful paired, multi-seed comparison—absent from most prior pipelines—tells a different story.

Implications. The contribution of this paper is therefore methodological and cautionary. The selection-bias-free protocol of Section III is inexpensive, transferable, and, in our case, decisive: it converts an apparently strong multimodal result—an average Sharpe ratio above two under test-set selection (Table II)—into an honest finding that, here, multimodal sentiment fusion does not help. We do not claim sentiment is universally useless—higher-frequency text, event-level rather than daily aggregation, or richer representations than a scalar may yet carry signal—but we do claim that any such result must be demonstrated under an evaluation protocol that controls for selection. We recommend that multimodal-finance studies (i) report a validation-locked technical-only baseline, (ii) report paired, multi-seed significance tests rather than per-stock maxima, and (iii) accompany Sharpe ratios with the Deflated Sharpe Ratio for the number of trials conducted [6], [7].

VII. LIMITATIONS

Our study is deliberately a focused re-evaluation, and its scope bounds its claims. (i) **Universe:** five large-cap U.S. equities and a single test year (2024); the negative result may not transfer to other regimes, small caps, or other markets, though the selection-bias mechanism is universal. (ii) **Sentiment representation:** a single daily FinBERT-Tone scalar per source; a richer representation (document embeddings, event tags, intraday text) could carry signal that scalar aggregation discards—testing this is the most promising direction for future work. (iii) **Data quality:** SPY news coverage in the

TABLE VIII
TEXT COVERAGE BY SPLIT. TRAIN: 2020–2022; VAL: 2023; TEST: 2024.

Stock	Split	News cov.	Social cov.	Mean S^{news}
AAPL	train	0.988	1.000	0.489
	val	1.000	0.992	0.466
	test	1.000	0.988	0.350
META	train	0.938	0.897	0.542
	val	0.968	0.836	0.447
	test	0.900	0.908	0.537
NVDA	train	0.896	1.000	0.482
	val	1.000	0.972	0.478
	test	1.000	1.000	0.359
SPY	train	0.927	0.918	0.393
	val	0.984	0.984	0.261
	test	0.406	1.000	0.486
TSLA	train	1.000	1.000	0.513
	val	1.000	0.968	0.435
	test	1.000	0.988	0.392

test year is only 41%, so SPY’s sentiment ablations are partly imputed; the conclusion is unchanged when SPY is excluded. (iv) **Foundation models** were run single-seed under compute limits; we therefore state their result as a directional finding, not a precise estimate.

VIII. CONCLUSION

We re-evaluated the reported benefits of multimodal sentiment fusion for short-term stock-movement prediction under a selection-bias-free protocol: an explicit chronological split with embargo, validation-locked model and horizon selection, multi-seed training, and paired significance testing. Across 4,200 task-specific runs and two time-series foundation models, the reported advantage of fusion did not survive. Test-set selection alone inflates the average Sharpe ratio from 1.40 to 2.29; under honest evaluation, no sentiment configuration—including a learned reliability-weighted fusion that generalizes the fixed 70:30 heuristic—beats a technical-indicator-only baseline, and foundation models do not beat task-specific models. The practical recommendation is not to abandon textual data, but to abandon optimistic evaluation: the value of a multimodal pipeline must be demonstrated against a validation-locked baseline with paired, multi-seed statistics. We release the protocol, code, and processed benchmark to make that the path of least resistance for future work.

REPRODUCIBILITY

All code, configuration files, processed per-stock features, and the full results table are released at GitHub, together with the immutable validation-locked selection file that all reported test numbers are read through. A reproducibility checklist accompanies the submission.

APPENDIX A PER-SPLIT DATASET STATISTICS

Table VIII reports text coverage and mean news sentiment separately for the training, validation, and test windows. The

notable irregularity is SPY news coverage in the test year (0.41), discussed in Section VII.

REFERENCES

- [1] Y. Xu and S. B. Cohen, “Stock movement prediction from tweets and historical prices,” in *Proc. 56th Annual Meeting of the Association for Computational Linguistics (ACL)*, 2018, pp. 1970–1979.
- [2] R. Sawhney, S. Agarwal, A. Wadhwa, and R. R. Shah, “Deep attentive learning for stock movement prediction from social media text and company correlations,” in *Proc. 2020 Conf. on Empirical Methods in Natural Language Processing (EMNLP)*, 2020, pp. 8415–8426.
- [3] K. Karadaş *et al.*, “Multimodal stock price prediction with news, social media and financial indicators,” *Expert Systems with Applications*, 2025.
- [4] Z. Wang *et al.*, “A learning-based multimodal approach to stock trend prediction with technical indicators and social media sentiment,” *IEEE Access*, vol. 11, 2023.
- [5] K. Kirtac and G. Germano, “Sentiment trading with large language models,” *Finance Research Letters*, vol. 62, p. 105227, 2024.
- [6] D. H. Bailey and M. López de Prado, “The deflated Sharpe ratio: Correcting for selection bias, backtest overfitting, and non-normality,” *The Journal of Portfolio Management*, vol. 40, no. 5, pp. 94–107, 2014.
- [7] D. H. Bailey, J. Borwein, M. López de Prado, and Q. J. Zhu, “The probability of backtest overfitting,” *Journal of Computational Finance*, vol. 20, no. 4, pp. 39–69, 2016.
- [8] M. López de Prado, *Advances in Financial Machine Learning*. John Wiley & Sons, 2018.
- [9] R. Arnott, C. R. Harvey, and H. Markowitz, “A backtesting protocol in the era of machine learning,” *The Journal of Financial Data Science*, vol. 1, no. 1, pp. 64–74, 2019.
- [10] C. R. Harvey, Y. Liu, and H. Zhu, “... and the cross-section of expected returns,” *The Review of Financial Studies*, vol. 29, no. 1, pp. 5–68, 2016.
- [11] S. Makridakis, E. Spiliotis, and V. Assimakopoulos, “Statistical and machine learning forecasting methods: Concerns and ways forward,” *PLOS ONE*, vol. 13, no. 3, p. e0194889, 2018.
- [12] K. Nyakurukwa and Y. Seetharam, “The news and social media sentiment–return relationship: Evidence from a frontier market,” *Journal of Behavioral and Experimental Finance*, vol. 41, p. 100894, 2024.
- [13] M. Jaggi, P. Mandal, S. Narang, U. Naseem, and M. Khushi, “Text mining of StockTwits data for predicting stock prices,” *Applied System Innovation*, vol. 4, no. 1, p. 13, 2021.
- [14] D. Araci, “FinBERT: Financial sentiment analysis with pre-trained language models,” *arXiv preprint arXiv:1908.10063*, 2019.
- [15] A. H. Huang, H. Wang, and Y. Yang, “FinBERT: A large language model for extracting information from financial text,” *Contemporary Accounting Research*, vol. 40, no. 2, pp. 806–841, 2022.
- [16] A. F. Ansari *et al.*, “Chronos-2: From univariate to universal forecasting,” *arXiv preprint arXiv:2510.15821*, 2025.
- [17] Z. Zhu *et al.*, “FinCast: A foundation model for financial time-series forecasting,” *arXiv preprint arXiv:2508.19609*, 2025.
- [18] S. Gu, B. Kelly, and D. Xiu, “Empirical asset pricing via machine learning,” *The Review of Financial Studies*, vol. 33, no. 5, pp. 2223–2273, 2020.
- [19] W. F. Sharpe, “Mutual fund performance,” *The Journal of Business*, vol. 39, no. 1, pp. 119–138, 1966.
- [20] R. Aroussi, “yfinance: Yahoo! finance market data downloader,” <https://pypi.org/project/yfinance/>, 2023.
- [21] L. Breiman, “Random forests,” *Machine Learning*, vol. 45, no. 1, pp. 5–32, 2001.
- [22] T. Chen and C. Guestrin, “XGBoost: A scalable tree boosting system,” in *Proc. 22nd ACM SIGKDD Int. Conf. on Knowledge Discovery and Data Mining*, 2016, pp. 785–794.
- [23] G. Ke *et al.*, “LightGBM: A highly efficient gradient boosting decision tree,” in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 30, 2017.
- [24] S. Hochreiter and J. Schmidhuber, “Long short-term memory,” *Neural Computation*, vol. 9, no. 8, pp. 1735–1780, 1997.
- [25] T. Head *et al.*, “scikit-optimize: Sequential model-based optimization in Python,” <https://scikit-optimize.github.io/>, 2021.